GONZALO ESTEVAS-GUILMAIN MORÁN

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PERSONAL STATEMENT

Flight Control System Engineer at Akkodis for Airbus Defence and Space, with an academic specialisation in Flight Dynamics and Control. After completing a BSc in Aerospace Engineering and the professionally qualifying MSc in Aeronautical Engineering at Universidad Politécnica de Madrid, specialised further through the MSc in Aerospace Dynamics at Cranfield University, with an overall average of 82.8/100 (Individual Research Project: 82/100). Current work focuses on requirement-based verification and validation of flight-control and flight-management models in MATLAB/Simulink. MSc research developed a learning-based Gust Load Alleviation controller for a flexible aircraft; results selected for publication, manuscript in preparation. Strong interest in flight dynamics, flight-control-law development and model-based V&V for advanced aircraft.

PROFESSIONAL EXPERIENCE

AKKODIS for Airbus Defence and Space, Getafe, Madrid, Spain. August 2026 - Present

Flight Control System Engineer | Flight Control Systems - Manage Flight & Autopilot

- Perform requirement-based verification and validation of MATLAB/Simulink models for Flight Control System (FCS), Flight Management System (FMS) and Independent Monitoring Flight Termination Unit (IMFTU) functions within military/UAS programmes (ATA 22/27).
- Design and execute tests in Simulink Test Manager and dedicated test harnesses, translating DOORS requirements into test inputs, expected results, pass/fail criteria and test sequences.
- Review model behaviour and investigate failed tests or requirement discrepancies, supporting model corrections when verification identifies implementation errors.
- Work in a collaborative configuration-controlled environment using Git and Bitbucket branches, coordinating technical activities directly with Airbus Defence and Space engineers.

AIRBUS. Xplorer programme - Structures and Structural Analysis, Madrid, Spain. February 2024 - July 2024

Structural Analysis Internship

- Developed and extended an analytical structural-analysis method through engineering programming in Fortran.
- Verified and implemented the method with Nastran finite element analysis, achieving 98% accuracy against reference results.
- Automated model creation tasks with TCL scripting, reducing setup time by more than two minutes per simulation.

EDUCATION

MSc in Aerospace Dynamics, Cranfield University, Cranfield, UK. October 2025 - September 2026

Specialisation in Flight Dynamics and Control. Overall average: 82.8/100 (taught modules 83.4/100; Individual Research Project 82/100); marks subject to Board of Examiners ratification.

Relevant modules: Flight Dynamic Principles; Flying Qualities and Flight Control; Air Vehicle Modelling and Simulation; Aircraft System Identification; Control Systems; Multivariable Control Systems; Flight Experimental Methods.

MSc in Aeronautical Engineering, Universidad Politécnica de Madrid, Madrid, Spain. September 2024 - September 2026

Professionally qualifying Master's degree; first-year average 7.5/10. Relevant coursework includes Flight Dynamics, Advanced Aeroelasticity, Optimisation & Control Techniques, Air Navigation Systems, Aerospace Electronic Systems, Advanced Structures and Space Vehicle Design/Certification.

Degree in Aerospace Engineering, Universidad Politécnica de Madrid, Madrid, Spain. September 2020 - July 2024

Aerospace Vehicles specialisation; GPA 7.5/10. Graduated with honours in the End-of-Degree Thesis and Computer Science. Relevant coursework included flight mechanics, fixed-wing and rotary-wing aircraft, missiles, automatic control, aerospace electronic systems and CAD/engineering design.

SELECTED TECHNICAL PROJECTS

Individual Research Project - Gust Load Alleviation System for Flexible BWB Aircraft

- Developed a gust-conditioned feedforward controller for a flexible blended-wing-body aircraft, replacing a fixed feedforward command with a case-dependent control strategy.
- Built an offline teacher/student framework in MATLAB using constrained optimisation (fmincon) to generate elevator/spoiler command histories and supervised neural networks to learn the gust/fuel scheduling map.
- Integrated and validated the frozen controller in nonlinear Simulink under actuator constraints and multiple gust/fuel conditions. Awarded 82/100. Results selected for publication; manuscript in preparation.

Flight Control Laws, Flying Qualities and Aircraft Modelling - Cranfield University

- Built nonlinear MATLAB/Simulink 6-DoF aircraft models and performed trim, linearisation, state-space/modal analysis and comparison with flight-test evidence.
- Designed SISO/MIMO control solutions using root locus, frequency response, PID/PD, state feedback, LQR/LQG concepts, gain scheduling and robustness-oriented analysis.
- Developed a gain-scheduled longitudinal stability-augmentation system with q-feedback, command pre-filters, bandwidth/PIO assessment and closed-loop verification.

Flight Test Data and Aircraft System Identification - Cranfield University

- Worked with Saab 340B flight-test data for kinematic-consistency checks, sensor bias/scale-factor correction and validation of measured aircraft responses.
- Applied Equation-Error and Output-Error identification methods to estimate dynamic behaviour and correlate identified models with flight-test data.

TECHNICAL SKILLS

- **Programming & Simulation:** MATLAB, Simulink, Python, Fortran, TCL, basic C++ fundamentals
- **Model-Based V&V & Requirements:** Simulink Test Manager, test harnesses, DOORS, requirement-based testing, Git and Bitbucket
- **Flight Dynamics & Control:** nonlinear 6-DoF dynamics, trim, linearisation, state-space models, stability modes, flying qualities, SISO/MIMO control, q-feedback, gain scheduling, LQR/LQG and robustness analysis
- **Flight Test & System Identification:** kinematic consistency, sensor error correction, Equation-Error/Output-Error methods, flight-test correlation and model validation
- **Engineering Software & CAD:** Nastran, HyperWorks, CATIA, Patran, Tecplot
- **Languages:** Spanish and Basque (native), English (C1), French (A1)

ADDITIONAL EXPERIENCE

Colegio Mayor Pío XII - Fundación Pablo VI, Madrid, Spain. September 2022 - July 2023

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- Coordinated day-to-day residential activities for approximately 100 students and helped organise events for more than 300 attendees, strengthening leadership, planning and team coordination.
- Full driving licence.